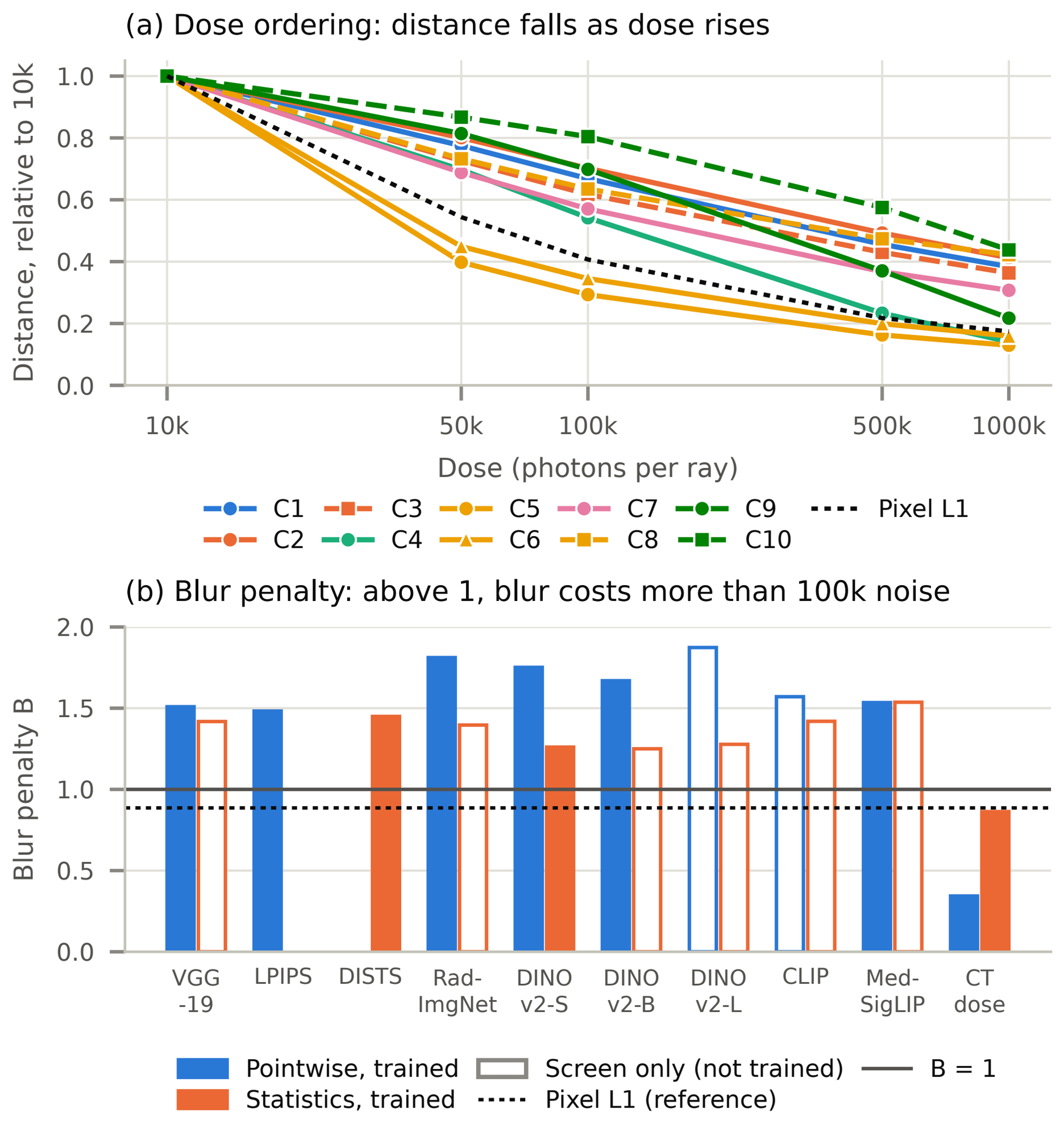
(a) Dose ordering: distance falls as dose rises
Distance, relative to 10k
1.0
0.8
0.6
0.4
0.2
0.0
10k
50k
100k
500k
1000k
Dose (photons per ray)
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
Pixel L1
(b) Blur penalty: above 1, blur costs more than 100k noise
Blur penalty B
2.0
1.5
1.0
0.5
0.0
VGG -19
LPIPS
DISTS
Rad- ImgNet
DINO v2-S
DINO v2-B
DINO v2-L
CLIP
Med- SigLIP
CT dose
Pointwise, trained
Statistics, trained
Screen only (not trained)
Pixel L1 (reference)
B = 1